S4 Experimental NMR details and ^{15}N solid-state NMR spectra

Experimental details for the ^{13}C solid-state CPMAS NMR spectra may be found in the original publication.[1] ^{15}N solid-state NMR spectra were recorded on a Bruker NMR spectrometer at a Larmor frequency of 30.42 MHz operating at a static magnetic field of 7.4 T (300 MHz). Samples were filled in a 4 mm ZrO_2 rotor and spun at 6 kHz. A standard cross-polarization MAS (CPMAS) experiment was done with a ^{1}H $\pi/2$ pulse at 64 kHz and a ramped CP (100 to 70%) with a contact time of 7.5 ms. The RF strengths of the contact pulses for ^{1}H and ^{15}N during CP were 58 and 36 kHz, respectively. For TAPD-$(Me)_2$ and TAPD-$(OMe)_2$ COFs, low signal-to-noise datasets were recorded, accumulating 92160 transients for TAPD-$(Me)_2$ and 82944 transients for TAPD-$(OMe)_2$ with a recycle delay of 2 s. Additional datasets exhibiting a higher signal-to-noise ratio were recorded by accumulating 261120 and 467520 transients for TAPD-$(Me)_2$ and for TAPD-$(OMe)_2$ respectively, again with a recycle delay of 2 s. ^{1}H decoupling was achieved with SPINAL-64 decoupling at 30 kHz.[2] ^{15}N NMR spectra were referenced to the chemical shift of labelled $^{15}NH_4NO_3$ solid. Line broadening of 100 Hz was applied during signal processing, to average out and reduce the noise level in the spectra. The data both with and without line broadening applied are shown in Figure S1. A longer collection time significantly reduces the amount of noise. Artificial line broadening also helps to mitigate the strong variations due to noise, but can also lead to the loss of meaningful details in the spectrum.

S5 g_R model with no averaging across layers

As mentioned in the conclusion of the main text, for some systems, it may be advantageous to avoid averaging chemical shifts across different layers. For this reason, we have used the g_R model and doubled the number of classes, thereby grouping carbon atoms into a